

Nathan Au

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Education

Concordia University

Bachelor of Engineering – Software Engineering, Co-op

Montreal, QC

Sep. 2024 – May 2028 (expected)

Relevant Skills

Technical Languages: Python, TypeScript, JavaScript, SQL, Dart, HTML, CSS, C++, Java

Frameworks & Libraries: FastAPI, Flutter, React, Next.js, Pandas, Matplotlib, NumPy, Pytest, Hugging Face Transformers, OpenCV, Ultralytics YOLO, Google ADK, Pydantic, LangChain, LiveKit, Zod, Stripe, Twilio

Practices: OOP, REST APIs, Agile Scrum, Git Version Control, Unit & E2E Testing, ETL Data Pipelines, LLM Integration, Agentic Systems, Prompt Engineering, CI/CD, Database Design, System Design, Docker Containerization

Spoken Languages: English (Native), French (Professional – CECR DELF B2)

Experience

Software Engineering Intern

Dendrometric

May 2026 – Aug. 2026

Montreal, QC

- Incoming Summer 2026

Software Engineering Intern

Biotics Labs

Jan. 2026 – Apr. 2026

Montreal, QC

- Scaled a conversational Twilio SMS user onboarding pipeline that collects 30+ data points with async message processing for metrics calculation, context generation, and nutrition plan creation to eliminate production timeouts.
- Shipped end-to-end full-stack features from frontend UI (Next.js, shaden/ui) to backend API design (FastAPI), database schema and authentication (Supabase), payment processing (Stripe), and production CI/CD (Railway).
- Architected a LiveKit-based agentic voice system with tool-calling LLMs, custom SessionManagers for persistent multi-turn conversations, and database-integrated tools to autonomously recommend and collect meal orders.
- Deployed production LLM infrastructure using Gemini and LangChain with Langfuse prompt management and context aggregation from user profiles and conversation history to enable personalized interactions at scale.

Software Developer

Google Developer Group Montreal

Nov. 2024 – Apr. 2025

Montreal, QC

- Developed a cross-platform Flutter mobile application that helps university students in Montreal connect with peers, discover local study spots, and stay up-to-date on upcoming campus events all on one platform.
- Built a data-driven recommendation engine using user profiles managed in Firebase Realtime Database and Gemini LLM integration via Vertex AI to generate personalized connection suggestions based on shared interests and goals.
- Integrated the Concordia University Open Data API to retrieve 60+ upcoming campus events in real-time and implemented responsive geolocation-based features using Flutter Map for interactive study spot discovery.
- Delivered technical presentations at Flutter Montreal and Concordia University to audiences of 100+ attendees and produced the official launch video to showcase system architecture and data flow.

Projects

Reel Digest | Python, Telegram Bot API, Ollama (Granite 4), SQLite, yt-dlp, OpenCV | [Source Code](#) **Dec. 2025**

Messaging-based productivity tool that saves users' time by turning short-form content into concise text summaries.

- Designed a multi-modal media processing pipeline with Google STT for audio transcription and OpenCV computer vision/Tesseract OCR optical character recognition for on-screen text extraction to enable contextual LLM analysis.
- Configured a SQLite caching layer to eliminate redundant inference, persist user history, and successfully serve live summaries for 300+ unique production requests.

TeeFour AI | Python, FastAPI, Ollama (Gemma 3), SQLAlchemy, Tesseract OCR, Pytest | [Source Code](#) **Oct. 2025**

Automated tax document processing system that eliminates manual data entry for tax accounting workflows.

- Engineered a two-layer document classification algorithm using PyMuPDF/Tesseract OCR to categorize known document types (T4s, IDs, receipts) via filename analysis and content-keyword matching logic.
- Streamlined tax form data field extraction by integrating a locally hosted LLM to transform unstructured OCR text inputs into validated, machine-readable JSON outputs for downstream use in tax filing software.

More projects: nathan-au.github.io